

## Multi-Modal Agentic AI System for Medical Diagnosis and Triage Project Timeline

## 1. Project Overview

This document presents the full project timeline for the Multi-Modal Agentic AI System for Medical Diagnosis and Triage. The system autonomously diagnoses chest and cardiovascular conditions by combining three machine learning models, a medical large language model (MedGemma-27B-IT), and a retrieval-augmented generation (RAG) layer. The pipeline processes chest X-ray images, structured electronic health records (EHR), and free-text clinical notes, and delivers a triage decision with specialist referral recommendation.

## 2. Architecture Summary

The system comprises three ML models, one agent LLM, and one RAG layer:

| Component                | Technology / Model                          | Type                   | Purpose                                                 |
|--------------------------|---------------------------------------------|------------------------|---------------------------------------------------------|
| Model 1 — ChestVision    | ResNet-50 / ViT-B/16 (from scratch)         | Custom Neural Network  | Multi-label X-ray classification (14 conditions) → ONNX |
| Model 2 — ClinicalFusion | XGBoost + OpenMed NER encoder fusion        | Hybrid Model           | Diagnosis (5 categories) + Severity (0–3) → ONNX        |
| Model 3 — OpenMed NER    | OpenMed-NER-DiseaseDetectSuperClinical-434M | Pretrained + Finetuned | Biomedical NER on clinical notes → ONNX                 |
| Agent LLM                | google/medgemma-27b-it                      | Instruction-tuned LLM  | Final diagnosis reasoning via HuggingFace Inference API |
| RAG Layer                | A-Z Medical Encyclopedia + ChromaDB         | Post-decision QA       | Supporting context enrichment after diagnosis           |
| Application              | Next.js 14 + LangGraph.js + Clerk + NeonDB  | Full-stack app         | Patient-facing UI, agent orchestration, appointments    |

## 3. Phase-by-Phase Timeline

**Phase 1: Data Preparation** | 20 May – 2 Jun 2026 | Weeks 1 – 2

**Objective:** Establish clean, split data pipelines for all three models. No model training starts until data is verified and processed.

| Task                            | Dates     | Status   | Deliverable                           |
|---------------------------------|-----------|----------|---------------------------------------|
| NIH ChestX-ray14 Preprocessing  | 20–26 May | Complete | train/val/test CSVs (85,108 images)   |
| Synthea EHR Feature Engineering | 20–26 May | Complete | Scaled feature matrix + label encoder |

|                                       |              |         |                                                |
|---------------------------------------|--------------|---------|------------------------------------------------|
| BIO-tag NER Dataset (Synthea Notes)   | 27 May–2 Jun | Pending | BIO-tagged NER dataset for Model 3 fine-tuning |
| Medical Encyclopedia → ChromaDB Setup | 3–9 Jun      | Pending | Populated ChromaDB collection (RAG)            |

**Phase 2: Model 3 — OpenMed NER (Pretrained + Transfer Learning)** | 27 May – 16 Jun 2026 | Weeks 2 – 4

**Objective:** Fine-tune the OpenMed-NER-DiseaseDetect-SuperClinical-434M pretrained model on Synthea chest/cardiovascular notes. This model must be completed first as its encoder feeds directly into Model 2 training.

| Task                           | Dates        | Status  | Deliverable                                           |
|--------------------------------|--------------|---------|-------------------------------------------------------|
| OpenMed NER Fine-tuning        | 27 May–9 Jun | Pending | Fine-tuned NER weights on Synthea notes               |
| NER Evaluation (seqeval F1)    | 10–12 Jun    | Pending | Entity F1 $\geq 0.82$ ; transfer gain $\geq 3$ F1 pts |
| ONNX Export — NER Model (INT8) | 13–16 Jun    | Pending | Quantised ONNX model ~150MB                           |

**Phase 3: Model 2 — ClinicalFusion (Hybrid Model)** | 10 Jun – 1 Jul 2026 | Weeks 3 – 6

**Objective:** Build the hybrid XGBoost + OpenMed NER encoder fusion model trained on Synthea EHR data. Produces diagnosis classification (5 categories) and severity scoring (0–3).

| Task                                 | Dates        | Status  | Deliverable                                   |
|--------------------------------------|--------------|---------|-----------------------------------------------|
| XGBoost Tabular Baseline Training    | 10–17 Jun    | Pending | XGBoost baseline; GridSearchCV results        |
| Fusion Network + OpenMed Integration | 17 Jun–1 Jul | Pending | Trained fusion model; ablation table          |
| ClinicalFusion Evaluation & Ablation | 24–28 Jun    | Pending | Macro F1 $\geq 0.80$ ; false-low rate $< 2\%$ |
| ONNX Export — ClinicalFusion         | 29 Jun–1 Jul | Pending | Validated ONNX pipeline ~120MB                |

**Phase 4: Model 1 — ChestVision (Custom Neural Network)** | 17 Jun – 15 Jul 2026 | Weeks 4 – 8

**Objective:** Train a custom ResNet-50 (and optionally ViT-B/16) from scratch on NIH ChestX-ray14 for multi-label thoracic condition classification. GPU-intensive — scheduled to run overnight in parallel with application build.

| Task                             | Dates        | Status  | Deliverable                                      |
|----------------------------------|--------------|---------|--------------------------------------------------|
| ResNet-50 / ViT-B/16 Training    | 17 Jun–8 Jul | Pending | Best checkpoint; training curves; Optuna report  |
| Grad-CAM & Explainability        | 8–10 Jul     | Pending | Heatmap overlays; localisation AUC               |
| ChestVision Evaluation (AUC-ROC) | 8–11 Jul     | Pending | Mean AUC $\geq 0.82$ ; per-class AUC $\geq 0.75$ |

|                                              |           |         |                                              |
|----------------------------------------------|-----------|---------|----------------------------------------------|
| ONNX Export + INT8 Quantisation (All Models) | 11–15 Jul | Pending | All 3 models as validated ONNX files < 250MB |
|----------------------------------------------|-----------|---------|----------------------------------------------|

**Phase 5: Application Build**
| 17 Jun – 19 Jul 2026
| Weeks 5 – 9

**Objective:** Build the full-stack Next.js 14 application with Clerk authentication, NeonDB, LangGraph.js agent, and MedGemma-27B-IT integration. Runs in parallel with model training phases.

| Task                                 | Dates        | Status  | Deliverable                                     |
|--------------------------------------|--------------|---------|-------------------------------------------------|
| Next.js 14 + Clerk + NeonDB Scaffold | 17 Jun–1 Jul | Pending | Auth-gated app with DB schema and routes        |
| ONNX Inference API Routes (×3)       | 1–8 Jul      | Pending | 3 working model inference endpoints             |
| LangGraph Agent + MedGemma-27B-IT    | 8–15 Jul     | Pending | 11-node stateful diagnostic agent               |
| ChromaDB RAG Integration             | 8–12 Jul     | Pending | Top-3 encyclopedia retrieval wired to agent     |
| Google Calendar + Gmail API          | 12–19 Jul    | Pending | Appointment booking and email confirmation flow |

**Phase 6: Production Deployment**
| 15 – 22 Jul 2026
| Weeks 9 – 10

**Objective:** End-to-end testing and production deployment to Vercel. CI/CD pipeline via GitHub Actions. All ONNX models validated within Vercel 250MB function size limits.

| Task                                 | Dates     | Status  | Deliverable                                    |
|--------------------------------------|-----------|---------|------------------------------------------------|
| E2E Testing & QA (Playwright / Jest) | 15–19 Jul | Pending | Smoke test with synthetic patient scenario     |
| Vercel Deployment + CI/CD Pipeline   | 19–22 Jul | Pending | Live production URL; automated ONNX validation |

**Phase 7: Report Writing & Submission**
| 15 – 24 Jul 2026
| Weeks 9 – 10

**Objective:** Final report writing, presentation preparation, and project submission. Runs in parallel with production deployment.

| Task                               | Dates     | Status  | Deliverable                                            |
|------------------------------------|-----------|---------|--------------------------------------------------------|
| Report Writing & Presentation Prep | 15–24 Jul | Pending | Final report + presentation slides submitted by 24 Jul |

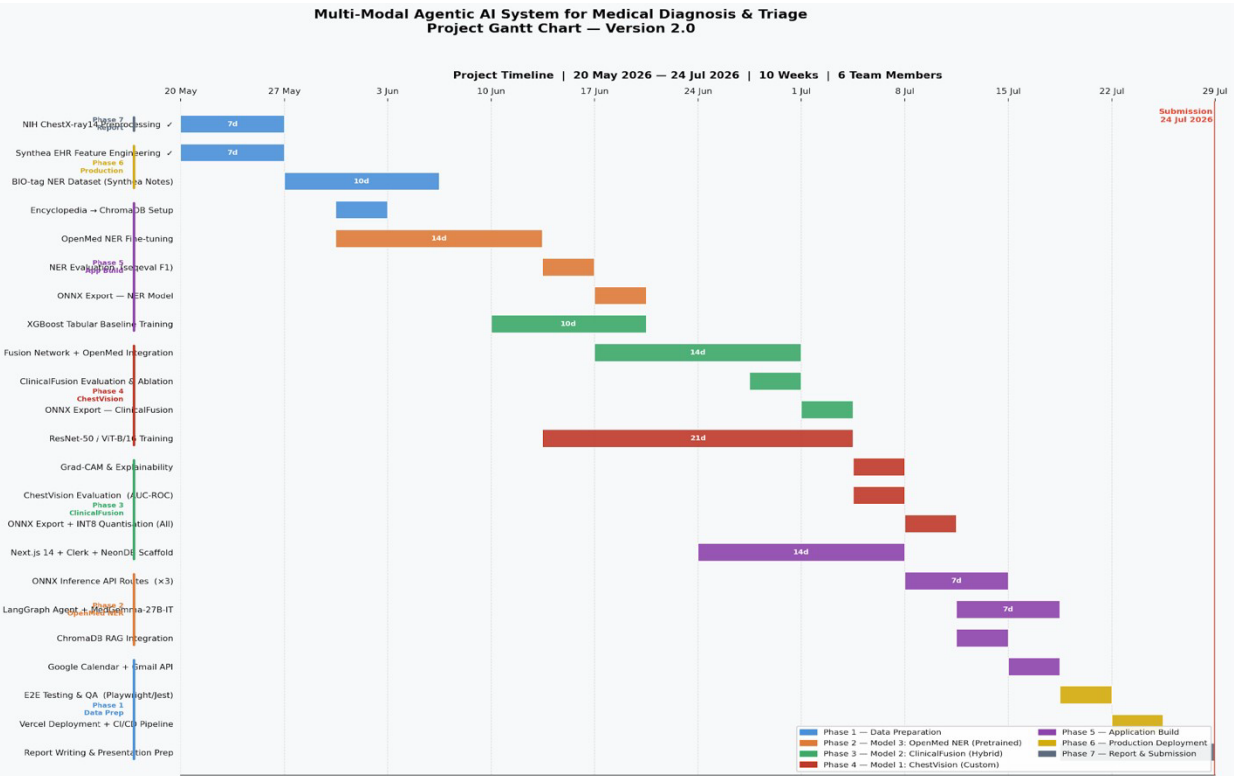
#### 4. Key Milestones

| Milestone | Target Date | Project Week |
|-----------|-------------|--------------|
|-----------|-------------|--------------|

|                                     |             |               |
|-------------------------------------|-------------|---------------|
| Phase 1 Complete                    | 2 Jun 2026  | End of Week 2 |
| OpenMed NER ONNX Export Ready       | 16 Jun 2026 | End of Week 4 |
| ClinicalFusion Model Validated      | 1 Jul 2026  | End of Week 6 |
| All 3 ONNX Models Quantised & Ready | 15 Jul 2026 | End of Week 8 |
| Full Agent End-to-End Test Passing  | 19 Jul 2026 | Week 9        |
| Production Deployment Live (Vercel) | 22 Jul 2026 | Week 10       |
| Final Report & Project Submission   | 24 Jul 2026 | Week 10       |

5. Project Gantt Chart

The Gantt chart below illustrates the full 10-week



schedule, showing task dependencies, parallel workstreams, and the production deployment phase immediately before the submission deadline.